



Karnatak Law Society's Gogte Institute Of Technology, Belagavi

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Department of Electronics and Communications Engineering

Project Phase 1

Batch No: **22ECPB20**

Project Type: “Oil skimming wireless boat to collect waste oil in ocean”

Project Type: Societal Application

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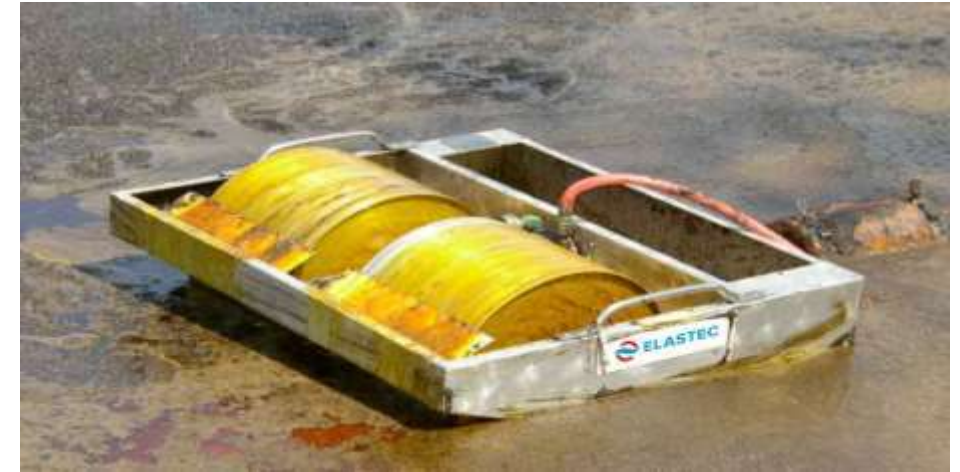
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*Papers need to be listed in descending order of published year

Introduction

- Oil pollution in oceans is a major environmental concern caused by oil spills from ships, industries, and offshore drilling activities. These pollutants severely affect marine life, water quality, and coastal ecosystems.
- Traditional oil spill cleanup methods are often **slow, labor-intensive, and risky** for human operators.
- To address this issue, this project proposes a **wireless oil skimming boat** that can **autonomously or remotely collect oil from the water surface**. The system uses a **floating mechanism and skimming technology** to separate oil from water efficiently.



Literature Survey

Ref. No	Title of the Paper	Publish ed Year	Journal / Conf Name	About the paper	Gaps identified
[1]	Study Of Operating Conditions For Oil Skimmer Apparatus From Water	2017	IEEE	This project focuses on efficient oil removal from water by optimizing operating conditions and analyzing Oil Recovery Rate (ORR) and Oil Recovery Efficiency (ORE) . It uses an ultra-mini oil skimmer with a rotating belt mechanism to collect oil, supported by temperature and pH measurement systems. Data is analyzed using the LINEST function and Least Square Method to develop accurate prediction models.	This study is primarily conducted under controlled laboratory conditions , which may not fully represent real-time environmental scenarios. The system lacks automation, smart sensing, and real-time monitoring capabilities , and does not integrate advanced technologies such as IoT or AI . Additionally, the analysis is limited to small-scale experiments
[2]	Research Progress and Prospects of Marine Oily Wastewater Treatment:	2019	MDPI	This study focuses on reviewing existing technologies for marine oily wastewater treatment , including physical, chemical, and biological methods, along with filtration, adsorption, and membrane separation techniques. The system uses oil-water separation units, filtration devices, membrane treatment systems, and adsorption materials to remove contaminants effectively. It is supported by analytical tools and simulation models to evaluate treatment efficiency and optimize the overall process.	However, several limitations exist, such as high cost and energy consumption , which make large-scale implementation challenging. The complex composition of marine oily wastewater makes treatment difficult, and many methods are less effective for emulsified oil removal . Additionally, the system lacks smart monitoring technologies and real-time marine deployment capabilities , limiting its practical application.
[3]	Sensors, Features, and Machine Learning for Oil Spill Detection and Monitoring: A Review	2020	MDPI	his study focuses on reviewing remote sensing sensors used for detecting oil spills in oceans and understanding feature extraction and selection techniques for accurate identification. It also analyzes machine learning and deep learning methods for automatic oil spill detection and monitoring. The system relies on hardware such as satellite sensors (Sentinel, Landsat, MODIS), SAR radar sensors, and optical/infrared sensors to collect ocean surface images. These are processed using machine learning algorithms (SVM, Random Forest, Neural Networks) along with image processing.	The system depends heavily on satellite data quality and availability , which may be affected by weather conditions, resolution, and noise. While SAR sensors work in all conditions, optical sensors are limited by cloud cover and lighting . Machine learning and deep learning models require large labeled datasets and high computational power , which can be challenging to obtain. Additionally, variations in ocean conditions may lead to false detections , and real-time monitoring and deployment are still limited in practical scenarios.

Ref. No	Title of the Paper	Publish ed Year	Journal / Conf Name	About the paper	Gaps identified
[4]	Marine Oil Spill Monitoring and Skimming Technology	2022	IEEE	This project focuses on developing an automated oil skimming system using IoT , where sensors are used to detect the presence of oil in water and automatically activate the skimming process. The system is built using hardware components such as Raspberry Pi 4, ultrasonic sensor (HC-SR04), MQ3 gas sensor, pH sensor, L298N motor driver, and a 12V DC motor . The operation and control are implemented using Python programming ,	The system is primarily tested as a prototype and is not yet suitable for large-scale deployment. It has limited intelligence , as it does not incorporate AI-based oil detection, and lacks autonomous navigation for operation in large water bodies. Additionally, the performance of the system may decrease under rough water conditions .
[5]	Detection of Oil Spill Events at Sea using Machine Learning	2023	IEEE	This study focuses on developing a machine learning model to predict oil spill events using AIS ship data by analyzing key ship parameters and estimating the probability of spills. It compares different algorithms such as Linear Regression, Decision Tree, and SVM to achieve accurate prediction results. The system uses hardware like satellite systems, drones/aircraft, and AIS ship monitoring systems .	The model's accuracy depends on the quality and availability of AIS and satellite data , which may not always be reliable. Machine learning algorithms require large and well-labeled datasets , and prediction performance may reduce with incomplete data. Additionally, real-time prediction requires high computational resources , and the system does not directly integrate with physical oil spill cleanup mechanisms.
[6]	Trash Collection And Oil Skimming Boat	2025	IEEE	This project aims to remove floating trash and oil spills from water bodies by developing an autonomous boat for pollution control . It uses object detection techniques to identify waste on the water surface and reduce human effort in cleaning lakes and rivers. The hardware includes Raspberry Pi 4 Model B, Pi Camera, L298N motor driver, and ultrasonic sensors for control, detection, and obstacle avoidance.	The system is mainly designed as a small-scale prototype and is not yet suitable for large ocean environments. It lacks advanced navigation and AI optimization , which limits efficiency in complex conditions. Performance may also be affected by lighting, water waves, and obstacles , and real-time large-scale deployment is still a challenge.

	Title of the Paper	Publish ed Year	Journal / Conf Name	About the paper	Gaps identified
[7]	Trends in oil spillmodelling	2025	MDPI	This study reviews the research trends in oil spill modelling , analyzing different models used to predict the movement of oil spills in oceans and identifying gaps for future improvements. It utilizes hardware such as satellite monitoring systems and ocean sensors to observe environmental parameters like wind, waves, and currents, along with high-performance computing systems for running simulations. The software includes oil spill simulation models such as GNOME, OSCAR, and OILMAP , along with statistical tools like R and RStudio and bibliometric analysis tools such as Bibliometrix for research evaluation.	This project aims to design an automated oil–water separation system that improves oil recovery efficiency while also monitoring and enhancing water quality. It incorporates pH sensing and oxygenation mechanisms , along with IoT-based remote monitoring. The hardware includes ATmega328P/Arduino controller, ESP32 module, pH sensor, DC motor, water pump, air pump, and an oleophilic belt with scraper mechanism , powered by a battery.
[8]	Advanced Aqua Source Electric Oil Skimmer with Oxygenation and pH monitoring	2025	IEEE	This paper on IoT-enhanced traffic light control systems can significantly contribute to your mini project, which involves creating a barrier to prevent vehicles from jumping red lights. The system's use of sensors and automated mechanisms can inspire your project's design, particularly in detecting vehicles and controlling the barrier. By integrating similar technologies, your project can effectively prevent accidents and improve road safety. The paper's insights on dynamic traffic signal adjustment can also enhance your project's functionality. By leveraging these concepts, you can develop a more efficient and responsive system for managing traffic flow and enforcing traffic rules.	The system mainly removes surface-level oil and may not perform effectively in rough or wavy water conditions . It has limited capability in handling emulsified oil , lacks AI-based detection and autonomous navigation , and has been tested only as a small-scale prototype , limiting its large-scale applicability

Field Survey & Data Collection Phase as part of literature survey

- **Survey & Data Collection:**

- What kind of field visits were conducted?

Preliminary **site observation of nearby water bodies** was carried out to understand pollution conditions and identify oil presence and floating waste.

- What specific data or information was collected during the survey?

Data included **visual assessment of oil spread, surface conditions, floating waste, and environmental parameters such as pH and temperature**, supported by reference datasets.

- What testing or measurements were performed on-site?

Basic observations and indicative measurements such as **water quality parameters (pH, temperature) and oil spread characteristics** were considered for system design.

- What tools or methods were used for data collection?

A combination of **visual inspection, sensor-based parameters, and analysis of existing datasets and research studies** was used.

Problem Statement and Objectives

Problem Statement:

Oil pollution in water bodies causes serious environmental damage. Existing oil removal methods are inefficient, time-consuming, and lack automation and real-time adaptability. Hence, there is a need for a cost-effective and automated system to efficiently remove oil from water surfaces.

Objectives:

- To design and develop an efficient oil skimming mechanism capable of separating oil from water surfaces with minimal water content and high recovery rate.
- To implement a low-cost and eco-friendly floating system that can operate in contaminated water bodies for continuous oil spill cleanup.
- To evaluate the performance of the system based on parameters such as oil collection efficiency, operating speed, and adaptability to different water conditions.

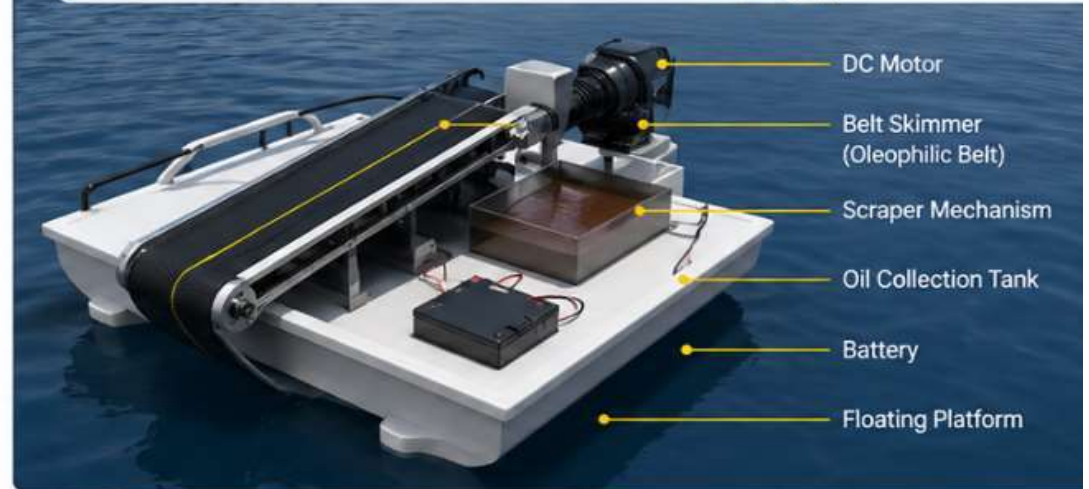
Plan for Project Deployment:

1 DETAILED ARCHITECTURE:

We propose an **Oil Skimming Boat System** designed to remove oil from water surfaces using a mechanical skimming mechanism. The system consists of:

- | | | |
|---|---|---|
|  | 1 FLOATING PLATFORM: | A lightweight floating structure supports all components and ensures stability on the water surface during operation. |
|  | 2 SKIMMER MECHANISM: | A belt-based skimmer made of oleophilic material is used to selectively attract and collect oil from the water surface. |
|  | 3 MOTOR DRIVE SYSTEM: | A DC motor drives the skimmer mechanism, enabling continuous rotation of the belt for efficient oil collection. |
|  | 4 OIL SEPARATION AND COLLECTION: | A scraper mechanism removes the collected oil from the belt and directs it into a storage container for separation and storage. |
|  | 5 POWER SUPPLY: | A battery provides the necessary power to operate the motor and other electrical components of the system. |
|  | 6 TWO-LEVEL VERIFICATION SYSTEM (ENHANCEMENT): | A sensor-based verification system ensures that the skimmer operates only in the presence of oil and confirms effective collection, improving overall efficiency. |
|  | 7 OUTPUT: | The system collects oil into a storage tank while leaving relatively clean water behind. |

OIL SKIMMING BOAT – OVERVIEW



WORKING FLOW



TWO-LEVEL VERIFICATION SYSTEM



Project Implementation Details



A smart oil skimming boat designed to remove oil from water surfaces using a mechanical skimming mechanism with efficient collection and storage.



HARDWARE USED



DC Motor

Drives the belt skimmer mechanism



Belt Skimmer

Oleophilic belt used to attract and collect oil



Battery

Provides power to the system components



Floating Platform

Provides buoyancy and supports all components



Oil Collection Tank

Stores the collected oil separately



(Optional) Arduino & Sensors

For automation, detection and control



PARAMETERS CONSIDERED



Oil thickness on water surface

Affects collection efficiency



Motor speed

Optimized for maximum oil recovery



Belt material

Oleophilic property ensures better oil adhesion



Stability of platform

Ensures smooth operation in water



IMPLEMENTATION PROCESS

1



Design and fabrication

Design of floating platform and system components

2



Installation

Installation of belt skimmer mechanism and scraper

3



Motor integration

Integration of DC motor for belt rotation

4



Oil collection setup

Setup of oil collection and storage unit

5



Testing and evaluation

Testing in controlled oil-water environment and result analysis

EXPECTED OUTCOMES



Design and development of a functional **wireless oil skimming boat prototype**.



Efficient separation and collection of **oil from water surfaces** with high recovery and minimal water content.



Achievement of **minimum 95% oil recovery efficiency** under controlled test conditions.



Reduction in manual intervention, ensuring **safer** and **faster** oil spill cleanup.



Development of a **low-cost, eco-friendly system** suitable for small-scale water bodies.



Stable and reliable operation under **varying surface conditions** (waves and disturbances).

FINAL TARGET OUTCOME



To achieve at least

95%

oil recovery efficiency

using the developed oil skimming system during testing and validation.



Outcomes of literature survey

Sl no	Type of Outcome	Completed? (Yes/No)
1	Survey paper	yes
2	Timeline preparation with respect to your project	Yes
3	Project design	yes
4	Flowchart	yes

Project work target :

- Publication/participation in State/National/International conference.
- Participation in Smart India Hackathon
- Uploading on Yukti portal and participation

*Papers need to be listed in descending order of published year

Timeline

Project work	May	June	July	August	Sept	Oct	Nov	Dec	Jan
System Planning & Resource Finalization	✓								
Literature Review & Design Validation	✓	✓							
Setup & Initial Implementation		✓	✓						
Objective 1 Development			✓	✓					
Objective 2 Development				✓	✓				
Objective 3 Development						✓			
Optimization & Documentation							✓	✓	
Project Outcome & Showcase									✓